Resolution Geometry: The Tensor Derivation

From Records to $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$
Standalone print: the full chain, its gates, and its destructors

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Abstract

This print derives the Einstein source and response from operational records in one chain, and it states exactly where geometry enters. The chain has four stages.

1. Records, before any spacetime. Continuity is constructed from protocol probabilities alone. The same construction realizes an interval, a circle or a Cantor set, so it does not select a geometry. Dynamics survives the quotient exactly when it respects future predictions. A predictive metric makes a total update Lipschitz, so it extends to the completion. Folds, waiting-time walls and zero-support conditioning are walls that continuity cannot cross on its own.

2. Balance, before any geometry. Balance holds on graphs, and it survives refinement through chain maps.

3. Geometry enters only through declared gates. A realization gate supplies the cut-flux form, with explicit counterexamples showing what fails without it. A quadratic-realization gate supplies a nondegenerate form of *unspecified* signature (p, q). Signature and dimension are then derived, not declared:

- predictive closure (a Lipschitz update) forces $\min(p, q) = 1$, because definite and ultrahyperbolic realizations are ill-posed;
- the mediator propagates and carries exchange between separated sources only for $D \geq 4$: there are $D(D-3)/2$ polarizations, the Weyl tensor vanishes identically in $D = 3$, and the static force is zero there;
- bound record carriers (orbits, atoms, clocks) are stable only for $D \leq 4$.

So the resolved realization is $(1, 3)$.

4. The tensor is forced. Local relabeling changes every tested matter Lagrangian by $\int T^{\mu}{}_{\nu} \partial_{\mu} \xi^{\nu}$ off shell, and T is exactly the metric-variation tensor. It follows that:

- the exchange index is momentum;
- the source is symmetric and universally coupled;
- the source is unique up to one constant per scalar bilinear;
- the compensator of relabeling is one symmetric tensor, which is the record metric;
- the conservation identity holds with the Levi-Civita connection.

The linear response is $G^{(1)}$ in every dimension. Self-coupling completes it to Einstein's equation, and the derived $D = 4$ makes the second-order completion unique (Lovelock).

Corrections. Totally antisymmetric torsion is invisible to every bosonic and point-particle record. It is therefore quotiented by descent, not excluded by the conservation identity, as an earlier print edition claimed. The first edition imported $D = 4$ and the Lorentz cone; this edition derives both from stated premises.

Evidence. One battery of 35 groups, symbolic or seeded, reruns byte-identically. Each group carries a canary.

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1 How to read this print

Status labels.

- CONSTRUCTED: proved from what precedes it.
- DESTRUCTOR LOCK: every alternative in a declared class is eliminated.
- GATE: a declared condition, shown with its failure modes.
- IMPORTED: a published theorem, cited.

- ANCHOR: a constant fixed by one measurement.
- OPEN: not established.

Every result names its battery group, e.g. [C3].

The chain.

	Level	Status	Groups
0	Records, protocol probabilities	definition	—
1	Operational continuity; descent; continuation	CONSTRUCTED	C1–C6
2	Balance on graphs; refinement	CONSTRUCTED	B1–B3
R	Realization: smooth oriented continuum, first-variation control, admissibility	GATE	R1–R3
B	Quadratic local realization, signature (p, q)	GATE	—
L	Lorentzian signature $\min(p, q) = 1$	CONSTRUCTED (predictive closure)	X3, D3
3	Relabeling response = source tensor	CONSTRUCTED	S1–S4
4	Universality	CONSTRUCTED	U1
5	Mediator = one symmetric tensor = record metric	CONSTRUCTED	M1
T	Transport	CONSTRUCTED + GATE	L1–L3
6	Linear response = $G^{(1)}$	CONSTRUCTED	E1
7	Nonlinear completion	CONSTRUCTED + IMPORTED	N1, N2
8	Dimension $D = 4$	CONSTRUCTED relative to Premises III–IV	X1, X2, X4, D2
9	Selection of spin and derivative order	DESTRUCTOR LOCK + data	D1, D3–D5
W	Lattice witness	executed	Q1, Q2

2 Records and operational continuity

Definition 2.1 (Records and protocols)**.** Let X be a set of record states and let $f_k : X \to [0, 1]$, $k \in \mathbb{N}$, be the probabilities of the outcomes of finite protocols. No coordinates, manifold, dimension or metric are assumed. Set $x \sim y$ iff $f_k(x) = f_k(y)$ for all k, and $Q = X/\!\sim$.

Proposition 2.2 (Operational continuity)**.** *[C1]* $d_{\mathrm{obs}}([x], [y]) = \sum_k 2^{-k} |f_k(x) - f_k(y)|$ *is a metric on* Q*. Its topology is the coarsest one in which every* f_k *is continuous.*

Proof. Symmetry and nonnegativity are clear. The triangle inequality holds termwise. $d_{\mathrm{obs}} = 0$ iff all f_k agree, i.e. iff the classes coincide. Each f_k is continuous, since $|f_k(x) - f_k(y)| \le 2^k d_{\mathrm{obs}}$. Conversely, an d_{obs}-ball contains a finite intersection of f_k-preimages, because the tail $\sum_{k>K} 2^{-k}$ is arbitrarily small. □

Proposition 2.3 (Non-selection)**.** *[C1] Every compact metrizable space* (K, ρ) *arises as a profile space. Choose a dense sequence* (a_k) *and set* $f_k = \rho(\cdot, a_k)/(1 + \rho(\cdot, a_k))$*.*

Proof. These are valid probabilities, and they separate points. The profile map $K \to [0,1]^{\mathbb{N}}$ is a continuous injection of a compact space into a Hausdorff space, hence a homeomorphism onto its image. □

Witness: an interval, a circle and a Cantor set are each realized injectively, with minimum profile separations 0.177, 0.501 and 0.023. *Consequence:* continuity built from records does not, and cannot, choose a manifold, a dimension or a metric tensor. Geometry needs a separate gate (Section 4).

Proposition 2.4 (Descent criterion)**.** *[C2] An update $U : X \to X$ induces $\bar{U} : Q \to Q$ iff $x \sim y \Rightarrow Ux \sim Uy$.*

Proof. Necessity: apply $\bar{U}$ to equal classes. Sufficiency: $\bar{U}[x] = [Ux]$ is then independent of the representative. □

Witness: in a seeded 14-state system, the present-observation quotient (3 classes) fails the criterion, because hidden information changes a future prediction. Its coarsest predictive refinement (11 classes) satisfies it.

Proposition 2.5 (Predictive metric and continuation)**.** *[C3, C4] Let U be total, let $d_0 \leq 1$ compare present observations, and let $0 < \beta < 1$. Define $d_\beta(x,y) = (1-\beta)\sum_{n\geq 0} \beta^n d_0(U^n x, U^n y)$. Then:*

(a) $d_\beta(Ux,Uy) = \big(d_\beta(x,y) - (1-\beta)d_0(x,y)\big)/\beta$, so U is β^{-1}-Lipschitz.

(b) $d_\beta(x,y) = 0$ iff $d_0(U^n x, U^n y) = 0$ for all n. This relation is the coarsest U-closed relation contained in present indistinguishability, so U descends to it.

(c) U extends uniquely to the metric completion.

Proof. (a) Shift the series by one. (b) The relation is U-closed, and any U-closed relation contained in $\{d_0 = 0\}$ is contained in it. (c) A Lipschitz map sends Cauchy sequences to Cauchy sequences; define $\hat{U}[x_n] = [Ux_n]$. Uniqueness follows from density and continuity. □

Witness: exact rational computation over all 196 pairs, with $\beta = 2/3$. The identity holds exactly, 76 pairs merged by the present observation are separated, and the zero set equals the predictive quotient of C2. *Canary (C4):* $\sin(1/x)$ is continuous on $(0,1]$, but two sequences tending to 0 have image limits $+1$ and -1. Without uniform control there is no extension.
Scope. The completion adds ideal predictive limits, not physically realized states. It gives no continuous time, proper time or cone.

Proposition 2.6 (Walls continuity does not cross)**.** *[C5]*

(a) Fold. *With $q = x^2$ and $\dot{x} = 1$, the states $\pm x$ share q but differ later by $4xt$. The quotient must be enlarged.*

(b) Waiting time. *$\dot{q} = 2\sqrt{q}$, $q(0) = 0$ has the C^1 solutions $q_\tau = \max(0, t-\tau)^2$ for every $\tau \geq 0$. A microscopic lift must select one.*

(c) Conditioning. *If $\mathrm{TV}(P,Q) = \delta$ and $P(Y), Q(Y) \geq \varepsilon$, then $\mathrm{TV}(P(\cdot|Y), Q(\cdot|Y)) \leq 2\delta/\varepsilon$. At vanishing support there is no control: the joint TV equals $\varepsilon \to 0$ while the conditional TV equals 1.*

Proof. (a) and (b) by direct computation. For (c), write $|P(A|Y) - Q(A|Y)| \leq |P(AY) - Q(AY)|/P(Y) + Q(A|Y)\,|Q(Y) - P(Y)|/P(Y) \leq 2\delta/\varepsilon$. For the converse, take two laws that differ only on a rare branch with disjoint outcomes. □

Witness: the bound holds on 2000 seeded random laws, with maximum ratio 0.5; the counterexample is exact for $\varepsilon = 10^{-1}, 10^{-2}, 10^{-3}$.

Proposition 2.7 (Chart crossing)**.** *[C6] On the circle take $z = \tan(\theta/2)$ and $w = \cot(\theta/2) = 1/z$. The equation $\dot{z} = (1+z^2)/2$ transports to $\dot{w} = -(1+w^2)/2$. At $\theta = \pi$, z diverges while $w = 0$ and the profile $(\cos\theta, \sin\theta)$ stays regular.*

A failed chart is crossed when the predictive state and the update survive in another representation. Globally this needs invertible overlaps, cocycle consistency and the Hausdorff property.

3 Balance before geometry

Theorem 3.1 (Cut balance)**.** *[B1] On an oriented graph with exchange J_e on the edge $e : i \to j$ and storage P_i, node balance $\Delta P_i = -(BJ)_i$ implies, for every region R,*

$$\Delta P_R + \sum_{e: i \to j} \big(\mathbf{1}_R(i) - \mathbf{1}_R(j)\big) J_e = 0.$$

Proof. Sum over $i \in R$. Internal edges cancel. □

Witness: 200 random regions with integer exchanges, all exact. No metric, distance or volume enters.

Proposition 3.2 (Refinement)**.** *[B2] Let fine and coarse incidences $B_{h'}, B_h$ be related by node aggregation R_0 and edge aggregation R_1, with $B_h R_1 = R_0 B_{h'}$. If $j_h = R_1 j_{h'}$ and $b_h = R_0 b_{h'}$, then $B_{h'} j_{h'} = b_{h'}$ implies $B_h j_h = b_h$.*

Proof. $B_h R_1 j_{h'} = R_0 B_{h'} j_{h'} = R_0 b_{h'}$. □

Witness: a 9-node graph aggregated into 4 clusters satisfies the chain map exactly. *Canary:* counting one internal edge breaks it.

Proposition 3.3 (Consistency and stability)**.** *[B3] Suppose $d(R_h F_h x, F R_h x) \le \epsilon_h$ and F is L-Lipschitz. Then the trajectory error obeys $e_k \le L^k e_0 + \epsilon_h \sum_{j<k} L^j$.*

Proof. $e_{k+1} \le \epsilon_h + L e_k$; induct. □

Canary: defects of size h accumulated over $1/h$ steps give an error of 1, so closure holds only on bounded windows.

4 The realization gate

Gate A (Realization)**.** (R1) The refinement family admits an oriented smooth d-dimensional realization. In it, aggregated exchange is the integral of a local, additive, orientation-odd flux density over cuts, and bulk exchange is bounded: a cell of size ϵ has net flux $O(\epsilon^d)$ and face fluxes $O(\epsilon^{d-1})$.

(R2) Sources converge as *first variations*: strong convergence of the fields entering quadratic sources, or an explicitly retained defect stress.

(R3) Admissibility at interfaces is inherited from the microscopic rule.

Proposition 2.3 shows that this gate is not implied by Sections 2–3. It is where geometry enters.

Proposition 4.1 (Cut-flux form under R1)**.** *[R1] The leading flux is linear in the area covector: $f_x(a)^A = \mathcal{J}^{\mu A}(x) a_\mu$, i.e. $\tau \in V \otimes \Lambda^{d-1} T^*M$. For a closed system, $d\tau = 0$ weakly.*

Proof. The face area covectors of a shrinking simplex sum to zero. Divide the balance by ϵ^{d-1}; the bulk term vanishes in the limit. Comparing the oblique face with the coordinate faces gives linearity (Cauchy's tetrahedron argument). □

Destructor: on 200 random tetrahedra, the linear law balances to 6×10^{-15}. Three nonlinear odd laws fail, with relative imbalances between 5.3 and 42: $|a|$, $a_1|a_1|$, and a linear law plus a_1^3.

Proposition 4.2 (Why R2 is needed)**.** *[R2]*

(a) $u_n = \sin(nx)/n \to 0$ *uniformly, yet* $\frac{1}{2}(u_n')^2$ *has mean* $\frac{1}{4}$ *for every* n*. The stress of the limit field is not the limit of the stresses.*

(b) $\|a_n \otimes a_n - a \otimes a\|_{L^1} \le (\|a_n\|_{L^2} + \|a\|_{L^2})\|a_n - a\|_{L^2}$*. Strong* L^2 *convergence of gradients controls quadratic sources.*

(c) $S_h(q) = h \sin(q/h^2) \to 0$ *uniformly, while* $S_h'(0) = 1/h$*. Convergence of action values does not control first variations.*

Proposition 4.3 (Why R3 is needed)**.** *[R3] Weak balance crosses interfaces through* $n_\mu[T^{\mu\nu}] = 0$*, but it does not select. For Burgers' equation with states* $-1 \,|\, +1$*, both the stationary shock and the rarefaction* $u = x/t$ *balance. The shock produces entropy* $+\frac{2}{3}$*, so it is inadmissible. The viscous rule implies* $\partial_t \frac{u^2}{2} + \partial_x \frac{u^3}{3} = \epsilon\, \partial_x(u u_x) - \epsilon u_x^2$*: the admissibility sign is derived from the microscopic rule, not chosen after the limit.*

5 Quadratic gate, signature, relabeling premises

Gate B (Quadratic local realization)**.** The local realization carries a nondegenerate quadratic form η of some signature (p, q), $p + q = D$, with p entries -1 and q entries $+1$. The dynamics of fields is second order, with η as principal symbol. Neither (p, q) nor D is declared.

Theorem 5.1 (Signature from predictive closure)**.** *[X3, D3] A predictive update on the realization must be Lipschitz (Proposition 2.5). For* $\sum_i s_i \partial_i^2 u = 0$*, a mode* $e^{\lambda x_e + ik\cdot x}$ *evolving along a direction* e *of sign* s_e *obeys*

$$\lambda^2 = s_e \sum_{i \neq e} s_i k_i^2.$$

Growth is bounded for all frequencies in some direction iff $\min(p, q) = 1$*. That is, the form is Lorentzian up to overall sign. Otherwise* λ *grows linearly with the frequency cutoff (Hadamard ill-posedness), and no Lipschitz predictive update exists:*

- *definite forms are elliptic (amplification* 5×10^{21} *at* $k = 50$*,* $t = 1$*);*
- *ultrahyperbolic forms* $(p, q \ge 2)$ *grow in every direction.*

Witness: *sampled growth over all* (p, q) *with* $p, q \le 4$*. The bounded set is exactly*

$$\{(1,1), (1,2), (1,3), (1,4), (2,1), (3,1), (4,1)\},$$

and for every other signature the growth rises from about 10 *to about* 100 *as the frequency cutoff rises from* 10 *to* 100*.*

Write $\eta = \mathrm{diag}(-1, +1, \ldots, +1)$ with $q = D - 1$ space directions. The value of D is derived in Section 12. Sections 6–10 hold for every $D \ge 3$: their proofs are dimension-free, Theorem 10.1 is re-verified for $D = 3, \ldots, 6$ (X1), and the explicit witnesses are computed in $D = 4$. *Data:* quadrupole radiation, and $-3 \times 10^{-15} \le \Delta c/c \le 7 \times 10^{-16}$ for gravitational waves [12]. *Limit (Q2):* a continuous-time local rule gives only an effective cone (Section 14).

Premise I (Coincidence-preserving local relabeling)**.** The choice of representative is made independently at each event, so relabeling $x \mapsto x + \xi(x)$ is local. It preserves every recorded coincidence, so one and the same ξ acts on every sector whose events the record protocol locates. It acts on fields by $\delta_\xi \psi = -\mathcal{L}_\xi \psi$.

Premise II (Matter class)**.** Locally, the descended matter dynamics is a Lorentz-covariant Lagrangian of tensor fields that admits a covariant extension to metric backgrounds. The explicit witnesses use first derivatives. Spinors are covered by imported results.

Premise III (Exchange at a distance)**.** The mediator carries exchange records (momentum) between spatially separated sources. This is the hub of Proposition 11.1 and of universality. It is supported by Newtonian attraction and by gravitational radiation.

Premise IV (Stable record carriers)**.** Records are kept and timed by bound, stable periodic systems, i.e. clocks: orbits and atoms held together by interactions whose mediators are massless (Gauss-law potentials). This is the operational basis of RG duration.

6 The source tensor

Theorem 6.1 (Relabeling response)**.** *[S1] Under Premises I and II,*

$$\delta_\xi \mathcal{L} = -\partial_\lambda(\xi^\lambda \mathcal{L}) + T^\mu{}_\nu \, \partial_\mu \xi^\nu \quad \textit{off shell}, \qquad T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\partial(\sqrt{-g}\,\mathcal{L}_g)}{\partial g^{\mu\nu}}\Big|_{g=\eta},$$

where $\mathcal{L}_g$ is the covariant extension.

Proof. $\sqrt{-g}\,\mathcal{L}_g$ is a scalar density. Transporting every field, the metric included, gives

$$\delta_\xi(\sqrt{-g}\,\mathcal{L}_g) = -\partial_\lambda(\xi^\lambda \sqrt{-g}\,\mathcal{L}_g).$$

At $g = \eta$ the metric part of the variation is $\delta g^{\mu\nu} = \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$, and it contributes $-\frac{1}{2} T_{\mu\nu} \delta g^{\mu\nu} = -T^\mu{}_\nu \partial_\mu \xi^\nu$. Subtracting this from the total leaves the matter part as stated.

Witnesses: the identity was computed with T obtained independently by metric variation. The residual is zero for generic functions of four variables in three cases: a scalar with arbitrary V, two scalars coupled by $\lambda\phi^2\chi^2$, and Maxwell with Lie transport of A_μ. □

Corollary 6.2 (Soldering; source = cut flux)**.** *Take ξ constant in a region and vanishing outside a collar. Then $\int T^\mu{}_\nu \partial_\mu \xi^\nu$ is the flux of $T^\mu{}_\nu \xi^\nu$ through the collar. So the quantity T carries across a cut is the response to* translation*. The transported index of the cut-flux form of Proposition 4.1 is a translation index, and the metric-variation source and the cut flux are one tensor, identically.*

Proposition 6.3 (Alternatives destroyed)**.** *[S2] Couple a symmetric field through $\frac{1}{2}\int hS$ with $\delta h = \partial\xi + \partial\xi$. The total action is invariant under the single relabeling iff $\partial_\mu (S - T)^{\mu\nu} \equiv 0$.*

- Destroyed *(seeded probes give nonzero residuals): $2T$; $T + \eta\,\mathrm{tr}T$; and the internal current $J^\mu n^\nu$ of a complex scalar.*
- Survives: *$T + (\eta\Box - \partial\partial)\phi^2$, as Proposition 6.5 requires.*

Proposition 6.4 (Symmetry and spin)**.** *[S3]*

(a) Translation-only transport ($\delta A_\nu = -\xi \cdot \partial A_\nu$, components carried but not rotated) generates the canonical tensor $T_c{}^a{}_l = F^{ab}\partial_l A_b + \delta^a_l \mathcal{L}$. It is non-symmetric and $U(1)$-dependent, and off shell

$$T_c{}^a{}_l - T_B{}^a{}_l = \partial_b(F^{ab} A_l) - (\partial_b F^{ab}) A_l.$$

(b) *Lie transport generates T_B, which is symmetric, gauge invariant and equal to the Hilbert tensor.*

(c) *A constant antisymmetric stress w^{ij} conserves momentum but drives angular momentum at rate $\partial_\lambda M^{\lambda ij} = 2w^{ij}$. Rotational relabeling forbids it unless a spin current is recorded.*

Reading. Symmetry is what relabeling does to fields that carry indices. Spinors require the Kosmann Lie derivative and the Belinfante–Rosenfeld construction [1, 2] (IMPORTED).

Proposition 6.5 (Improvement freedom)**.** *[S4]*

(a) *$\alpha\eta\Box X + \beta\partial\partial X$ is identically conserved iff $\beta = -\alpha$.*

(b) *It carries no total energy-momentum: $\Delta T^{00} = -\alpha\nabla^2 X$ and $\Delta T^{0i} = \alpha\,\partial_t\partial_i X$ are spatial divergences.*

(c) *It is the flat-space form of ξRX. With internal gauge invariance and operator dimension at most 4, the only non-minimal couplings are $\xi_{ij}R\phi_i\phi_j$:*

- *RA^2 and $R_{\mu\nu}A^\mu A^\nu$ break $U(1)$;*
- *$R\bar{\psi}\psi$ has dimension 5;*
- *RFF has dimension 6.*

The source is therefore unique up to one ANCHOR per scalar bilinear. This is exactly the ambiguity of covariant extension in Theorem 6.1.

7 Universality

Theorem 7.1 (One relabeling, one coupling)**.** *[U1] Let $g = \eta + \kappa h$, and couple each sector A located by the protocol through $\frac{c_A}{2}\int hT_A$. Off-shell invariance under the single relabeling of Premise I forces $c_A = \kappa$ for every A. This includes sectors that never interact with each other.*

Proof. The variation is $\int \sum_A (c_A/\kappa - 1)T_A^{\mu\nu}\partial_\mu\xi_\nu$. Off shell each $\partial T_A \neq 0$, and the sectors' fields are independent.

Witness: two free scalars with masses 1 and 2. Seeded probes give the unique solution $c_1 = c_2 = 1$. □

Proposition 7.2 (Hub)**.** *The mediator exchanges momentum with every coupled sector, with $f_A \propto c_A$. Mediator consistency then gives $c_A(c_A - c_h) = 0$. With Theorem 7.1, $c_A = c_h = \kappa$.*

Destructor 7.3 (Active and passive mass)**.** [U1] In a Newtonian pair with active/passive ratios r_A, r_B, total momentum drifts by $\leq 8 \times 10^{-16}$ when the ratios are equal. It drifts by 5.2×10^{-3} for $r_B/r_A = 1.01$ and by 5.2×10^{-2} for 1.1: the isolated pair self-accelerates.

Data.

- MICROSCOPE: $\eta(\mathrm{Ti}, \mathrm{Pt}) = (-1.5 \pm 2.3_{\mathrm{stat}} \pm 1.5_{\mathrm{syst}}) \times 10^{-15}$ [10];
- lunar laser ranging: active and passive mass equal for Al and Fe to 3.9×10^{-14} [11].

8 The mediator

Theorem 8.1 (Compensator)**.** *[M1] The relabeling residual $\int T^{\mu\nu}\partial_\mu\xi_\nu$ has 10 independent symmetric components. A compensating field needs a local, Lorentz-covariant gauge variation, linear in $\partial\xi$, that pairs with all 10. The ranks absorbed are:*

- *scalar ($\partial\cdot\xi$): 1;*
- *2-form ($\partial_{[\mu}\xi_{\nu]}$): 0;*
- *scalar plus 2-form: 1;*
- *vector: none, since there is no one-index term;*
- *symmetric rank-2 field ($\partial_{(\mu}\xi_{\nu)}$): 10.*

The compensator is therefore one symmetric tensor h, with $\delta h_{\mu\nu} = \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$.

Corollary 8.2 (The compensator is the record metric)**.** *Matter clocks and rulers respond to $\eta + \kappa h$ (Theorem 6.1 with $g = \eta + \kappa h$). The field that restores relabeling invariance is the field the record protocol measures.*

9 Transport

Proposition 9.1 (Koszul uniqueness)**.** *[L1] Metric compatibility and torsion-freeness determine the connection uniquely as the Christoffel symbols. Without torsion-freeness, metric-compatible connections leave $n^2(n-1)/2$ free contorsion components.*

Witness: at a random point in $n = 3$, with random metric and first derivatives, the 27 unknowns are fixed and equal the Christoffel symbols; dropping torsion-freeness leaves 9 free components.

Proposition 9.2 (Noether II)**.** *[L2] If the matter action is invariant under all relabelings on a curved realization, then $\nabla_\mu T^\mu{}_\nu = (\Box\phi - V')\partial_\nu\phi$ with the Levi-Civita connection. This was verified on a generic two-dimensional metric with three free functions.*

Weyl non-metricity and vector torsion change the identity. In two dimensions both change it by the same term, $2\partial_\lambda f\, T^\lambda{}_\nu - T^\lambda{}_\lambda\partial_\nu f$. Scalar records can therefore resolve both.

Proposition 9.3 (Totally antisymmetric torsion is null; correction)**.** *[L3] Let H be an arbitrary 3-form and set $K^\rho{}_{\mu\nu} = H^\rho{}_{\mu\nu}$. Then $\hat\Gamma = \Gamma_{LC} + K$ is metric compatible with nonzero torsion $2K$, yet for every symmetric T, scalar ϕ and velocity u:*

$$(\hat\nabla - \nabla)_\mu T^{\mu\nu} = 0, \qquad g^{\mu\nu}K^\rho{}_{\mu\nu} = 0, \qquad K^\rho{}_{\mu\nu}u^\mu u^\nu = 0.$$

It appears only in $[\hat\nabla_\mu, \hat\nabla_\nu]\phi = -2K^\rho{}_{\mu\nu}\partial_\rho\phi$, which no minimally coupled bosonic Lagrangian contains. Maxwell uses $F = dA$, which needs no connection.

Proof. Every displayed term contracts K on an antisymmetric pair against a symmetric object. This was verified symbolically for generic H and T. □

Corollary 9.4 (Transport by descent)**.** *For scalar, Maxwell and point-particle protocols, the H-sector changes no prediction. By Proposition 2.4 it is null and quotiented. The predictive transport is Levi-Civita.*

Gate C (Transport)**.**

- Spinor (axial) protocols resolve H; there it is an additional field (Einstein–Cartan contact), declared.
- Resolvable deformations (Weyl non-metricity, vector torsion) would be additional fields detectable by scalar propagation; they are declared absent.

10 Linear response

Theorem 10.1 (Selection of the response operator). *[E1] The general local, Lorentz-covariant, parity-even, two-derivative operator on symmetric h is*

$$\mathcal{E}_{\mu\nu} = A\Box h_{\mu\nu} + B(\partial_\mu v_\nu + \partial_\nu v_\mu) + C\partial_\mu\partial_\nu h + \eta_{\mu\nu}(D\,\partial\cdot v + E\Box h), \qquad v_\nu = \partial^\mu h_{\mu\nu}.$$

Three conditions are relevant:

- gauge *($\mathcal{E}[\partial\xi + \partial\xi] = 0$, relabeling);*
- balance *($\partial^\mu\mathcal{E}_{\mu\nu} \equiv 0$);*
- reciprocity *(self-adjointness).*

Then:

(a) gauge alone gives $B = -A$, $C = A$, $D = -E$;

(b) balance alone gives $B = -A$, $C = -E$, $D = A$;

(c) any two give $\mathcal{E} = -2A\,G^{(1)}_{\mu\nu}$;

(d) gauge forbids the mass terms $a_0 h + b_0\eta h$;

(e) linearized Ricci is gauge invariant but not divergence-free.

Proof. Substitute a generic gauge mode, or a generic h, built from arbitrary functions of four variables, and require every coefficient of every independent derivative to vanish. For reciprocity, compare $\langle k, \mathcal{E}h\rangle$ with $\langle \mathcal{E}k, h\rangle$ on the Fourier symbol over all basis pairs. The selected operator agrees componentwise with $-2A$ times the linearized Einstein tensor built from the linearized Riemann tensor. □

11 Nonlinear completion

Proposition 11.1 (The mediator carries momentum). *[N1] The Newtonian field stress $t^{ij} = (\partial^i\Phi\partial^j\Phi - \frac{1}{2}\delta^{ij}|\nabla\Phi|^2)/4\pi G$ satisfies $\partial_j t^{ij} = \rho\,\partial_i\Phi$ exactly, which balances the force on matter. Bound matter has $\partial_\mu T^{\mu\nu}_{\mathrm{m}} = f^\nu \neq 0$. The identically divergence-free response of Theorem 10.1 cannot equal it, so the mediator must enter its own source.*

Imported Theorem 11.2 (Self-coupling; one graviton; Lovelock). *(a) Order-by-order consistent self-coupling of the field of Theorem 10.1 reconstructs diffeomorphism invariance and the Einstein–Hilbert action [3, 4].*

(b) No ghost-free two-derivative cross-coupling of several massless spin-2 fields exists [5].

(c) In $D = 4$, the only symmetric divergence-free tensor built from g and its first two derivatives is $aG_{\mu\nu} + bg_{\mu\nu}$ [6].

Check (N2), FLRW minisuperspace with lapse:

- Gauss–Bonnet has identically vanishing Euler–Lagrange equations in $D = 4$ and second-order ones in $D = 5$;
- R gives second-order equations;
- R^2 gives fourth-order equations.

Theorem 11.3 (Field equation). *Under Gates A, B and C, Premises I and II, and the imported results,*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa\Big(T_{\mu\nu} + \sum\nolimits_{ij}\xi_{ij}\big[G_{\mu\nu}\phi_i\phi_j + (g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu)(\phi_i\phi_j)\big]\Big),$$

with $\nabla^\mu G_{\mu\nu} \equiv 0$ enforcing $\nabla^\mu T_{\mu\nu} = 0$. The constants κ, Λ and ξ_{ij} are anchors.

12 Dimension

The question is how many space directions the Lorentzian realization can have. Too few, and curvature cannot carry anything away from its sources. Too many, and nothing bound by the mediated forces stays together.

Proposition 12.1 (The selection holds in every dimension)**.** *[X1] For $D = 3, \dots, 6$ (Fourier symbol, seeded momenta):*

- *each of gauge invariance and balance alone leaves a 2-dimensional family;*
- *any two of gauge, balance and reciprocity leave exactly $G^{(1)}$.*

In $D = 2$ the Einstein symbol vanishes identically (norm 6×10^{-17}): there is no response at all.

Theorem 12.2 (Lower bound: propagation needs $D \geq 4$)**.** *[X2]*

(a) Polarizations. *On shell, $G^{(1)}$ leaves $D(D-3)/2$ physical polarizations (kernel minus the D gauge modes): $0, 2, 5, 9, 14$ for $D = 3, \dots, 7$.*

(b) Free curvature. *The Weyl tensor has $D^2(D^2-1)/12 - D(D+1)/2$ components, which is 0 in $D = 3$ and 10 in $D = 4$. A generic 3-D metric has Riemann tensor of order 1.26 but Weyl tensor $\leq 10^{-15}$: curvature exists only where the sources are.*

(c) Static exchange. *The static numerator $1 - 1/(D-2)$ vanishes in $D = 3$, so separated masses exert no force.*

(d) In $D = 2$, $G_{\mu\nu} \equiv 0$ although $R \neq 0$.

Under Premise III, $D \geq 4$.

Theorem 12.3 (Upper bound: stable carriers need $D \leq 4$)**.** *[X4, D2] Let $n = D - 1$ space dimensions carry Gauss-law potentials $\propto r^{-(n-2)}$.*

(a) Newtonian orbits: *circular orbits are stable only for $n = 3$ (D2).*

(b) Relativistic orbits: *the Schwarzschild–Tangherlini metric $f = 1 - (r_s/r)^{D-3}$ has stable circular timelike orbits for $D = 4$, and for no $D = 5, 6, 7$ at any angular momentum.*

(c) Quantum carriers: *for $H = -\Delta - g\, r^{-(n-2)}$, a Gaussian trial state of width s has energy $E(s) = \frac{n}{2s^2} - g\, \frac{\Gamma(1)}{\Gamma(n/2)} s^{-(n-2)}$.*

- *$n = 3$: there is an interior minimum $-0.212\, g^2$, above the exact hydrogen value $-g^2/4$.*
- *$n = 4$: the problem is scale invariant, $E = (2-g)/s^2$. There is no ground state, and the energy is unbounded below for $g > 2$.*
- *$n \geq 5$: the variational upper bound tends to $-\infty$, so H is unbounded below.*

Under Premise IV, $D \leq 4$.

Corollary 12.4 (The resolved realization is $(1, 3)$)**.** *Gate B, Theorem 5.1, Theorem 12.2 and Theorem 12.3 give signature $(1, 3)$. In $D = 4$ Lovelock's theorem makes the second-order completion unique, and Gauss–Bonnet is topological (N2).*

Scope. The corollary concerns the *resolved* dimensions. Compact directions smaller than every accessible protocol are in the observational kernel. Inverse-square tests resolve no extra extended direction down to $52\,\mu$m [9].

Reading. Curvature can be carried away from its sources only from three space directions upward. Bound record carriers stay together only up to three. Three space directions and one time are the only realization in which both hold.

13 Selection by destruction

Destructor 13.1 (Mediator table). [D1] The light-bending parameter is $\gamma = N(\text{mass}, \gamma)/N(\text{mass}, \text{mass}) - 1$, computed from exact propagator numerators.

Mediator	γ	Verdict
spin 0, coupled to the trace	-1	destroyed
spin 1, coupled to the energy current	—	destroyed (like energies repel)
massive spin 2 (Fierz–Pauli), $D = 4$	$1/2$	destroyed
massless spin 2, $D = 3$	—	no static force (X2)
massless spin 2, $D = 5, 6$	$1/2,\ 1/3$	destroyed (also Theorem 12.3)
massless spin 2, $D = 4$	1	survives

For massless spin 2, $\gamma = 1/(D-3)$. The Cassini 2σ window, $\gamma - 1 \in [-2.5, 6.7] \times 10^{-5}$ [8], leaves one survivor. A Brans–Dicke admixture needs $\omega \geq 39{,}998$.

Destructor 13.2 (Derivative order). [D3, D4, D5]

- The fourth-order Pais–Uhlenbeck energy is $-\frac{1}{2}(\omega_1^2 - \omega_2^2)(\omega_1^2 a_1^2 - \omega_2^2 a_2^2)$: its modes have opposite signs, i.e. a ghost. Weyl^2 adds a massive spin-2 ghost [7].
- The healthy extension $R + \alpha R^2 - 2\Lambda$ contains every vacuum Einstein metric exactly (D4). The witness is Schwarzschild–de Sitter, with Kretschmann scalar minus its de Sitter value equal to $48M^2/r^6$; the wrong-Λ canary fails.
- Its weak field (D5) is $\Psi, \Phi = -\frac{GM}{r}(1 \pm \frac{1}{3}e^{-mr})$ with $m = 1/\sqrt{6\alpha}$, giving $\gamma(r) = (3 - e^{-mr})/(3 + e^{-mr})$.
- Cassini requires a range below 1.09×10^8 m. Torsion balances exclude gravitational-strength Yukawa forces beyond $38.6\,\mu$m [9].

The data select the second-order Einstein branch at every tested scale.

14 Lattice witness

Proposition 14.1 (Momentum is scale-emergent). *[Q1] Take a continuous-time ϕ^4 chain on a ring.*

- *Energy is conserved, to* 6.9×10^{-13}.
- *Label-translation momentum is exactly conserved for linear dynamics (drift* $\leq 5.6 \times 10^{-14}$*).*
- *On-site nonlinearity breaks it at $O(a^2)$: the drift is* 2.09×10^{-3}, 5.36×10^{-4}, 1.35×10^{-4} *and* 3.37×10^{-5} *for $N = 16$, 32, 64 and 128, a slope of* -1.99.

Proposition 14.2 (Effective cone, not a front; correction). *[Q2] Consider $\ddot{u}_i = u_{i+1} - 2u_i + u_{i-1}$ with a unit initial displacement. The response at distance r is*

$$u_r(t) = J_{2r}(2t),$$

with leading term $t^{2r}/(2r)!$ (verified for $r = 1, \ldots, 6$). It is nonzero at every distance for every $t > 0$, so there is no strict causal front.

Outside the maximal group velocity the leakage is bounded by $t^{2r}/(2r)!$, which decays faster than any exponential. At $t = 10$ it is:

- 1.2×10^{-4} *at $r = 15$;*

- 9.9×10^{-10} *at* $r = 20$*;*
- 2.3×10^{-23} *at* $r = 30$*.*

The lattice integration matches the Bessel form to 6×10^{-13}*.*

Reading. A continuous-time lattice yields only an effective cone. A discrete-time local reversible rule does better: the Dirac quantum cellular automaton has an exact cone, with amplitude exactly zero outside $|x| \leq t$ (Quantum Gravity print).

15 Ledger

Step	Status	Basis
Operational continuity, non-selection	CONSTRUCTED	Props. 2.2, 2.3
Dynamics on the quotient; continuation	CONSTRUCTED	Props. 2.4, 2.5
Folds, waiting time, zero support	OBSTRUCTIONS	Prop. 2.6
Balance and refinement before geometry	CONSTRUCTED	Thm. 3.1, Prop. 3.2
Smooth realization, first variations, admissibility	GATE A	Props. 4.1–4.3
Quadratic realization	GATE B	—
Lorentzian signature	CONSTRUCTED (predictive closure)	Thm. 5.1; X3
$D = 4$ (resolved)	CONSTRUCTED relative to Premises III–IV	Thms. 12.2, 12.3
Coincidence-preserving local relabeling; matter class; exchange at a distance; stable carriers	PREMISES	I–IV
Source = relabeling response = Hilbert = cut flux	CONSTRUCTED	Thm. 6.1, Cor. 6.2
Symmetry	CONSTRUCTED	Prop. 6.4
Uniqueness up to ξ_{ij}	CONSTRUCTED	Prop. 6.5
Universal coupling	CONSTRUCTED + data	Thm. 7.1
One symmetric mediator = record metric	CONSTRUCTED	Thm. 8.1
Levi-Civita transport	CONSTRUCTED (descent) + GATE C	Props. 9.1–9.3
Linear response $G^{(1)}$, massless	CONSTRUCTED	Thm. 10.1
Self-coupling, one graviton, Lovelock	CONSTRUCTED + IMPORTED	Prop. 11.1; [3, 4, 5, 6]
Spin 2, second order	DESTRUCTOR LOCK + data	D1, D3–D5
κ, Λ, ξ_{ij}	ANCHORS	—
Spinor transport; torsion contact	IMPORTED / GATE C	[1, 2]
Gate A for RG's own microscopic rule	OPEN	—
Strict cone from a microscopic rule	CONSTRUCTED (discrete time)	Quantum Gravity print
Compact dimensions below resolution	observational kernel	—
Singular and topology-changing passages	OPEN	—
Quantum regime	NOT ADDRESSED	—

The supported statement. *Records determine an operational continuity that selects no geometry. Balance holds and refines without geometry. Once a smooth quadratic realization is admitted (Gates A and B), predictive closure forces Lorentzian signature, and propagation together with stable record carriers forces three resolved space directions. Local coincidence-preserving relabeling then forces the following: the source is the symmetric, universally coupled relabeling response, unique up to one constant per scalar bilinear; the mediator is the single symmetric compensator, which is the record metric; predictive transport is Levi-Civita, the remaining torsion being null by descent for bosonic records; and the response is linearized Einstein, completed by self-coupling to* $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$. *Every alternative mediator in the local, two-derivative, spin at most 2 class is destroyed by internal inconsistency or by data.*

16 Corrections relative to the previous print edition

- *Transport.* The earlier claim that Noether II excludes torsion was too strong. Totally antisymmetric torsion passes the scalar, symmetric-stress and autoparallel tests (Proposition 9.3). It is removed by descent for bosonic protocols, and declared in Gate C for spinors.
- *Cone.* The earlier claim of an emergent cone from the lattice dispersion was too strong. The continuous-time lattice has Bessel tails at every distance; only an effective cone is supported (Q2).
- *Signature and dimension.* The first edition declared the Lorentz cone and imported $D = 4$ (Lovelock and data). This edition derives the signature from predictive closure and $D = 4$ from propagation and stable carriers (Section 12). The remaining gate is the existence of a quadratic realization.
- *Continuum premise.* The single "continuum regime" premise is split into what is constructed (Sections 2–3) and what is gated (Gate A). Shrinking simplices and area covectors belong to the gate, not to the records.

17 Reproduction

- `rg_tensor_derivation.py`, with `rg_curv.py`, runs all 35 groups in about a minute. The seed is 20261110 and reruns are byte-identical. Group names follow the brackets in the text.
- *Limits.*
  - Theorem 6.1 is witnessed on three Lagrangian families; its proof covers every Lagrangian with a covariant extension.
  - M1 is exhaustive within local, first-derivative, Lorentz-covariant gauge terms.
  - L2 is two-dimensional.
  - N2 is minisuperspace.
  - X4's orbit scan covers $L \in [0.2, 30]$ and $r \leq 400\, r_s$; the quantum bounds are variational.
  - The data are quoted from the cited publications, and no data were fitted.

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